

SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26116**
- **Problem Statement Title-Urban
Mixed-Use Design Challenge**
- **Theme-Vertical Synergy**
- **PS Category- Software**
- **Team ID-SIH26-A0H-T505**
- **Team Name – Archi_talezz**



Urban Mixed-Use Design Challenge

EXPLANATION:

- The Ground and First Floors accommodate commercial spaces such as retail shops, cafés, offices, and co-working areas.
- The 2nd to 9th floors provide comfortable residential units with balconies, natural lighting, and ventilation.
- Separate commercial and residential entrances with a centralized lift and staircase core ensure efficient circulation, privacy, and safety.

HOW IT WORKS:

- The basement solves urban parking requirements and includes **EV charging** for future mobility.
- Commercial spaces activate the street and provide economic activity, while upper floors create a live-work environment.
- Natural ventilation, daylight, shading, and greenery improve occupant comfort and reduce dependence on artificial systems.

UNIQUENESS :

- The key innovation is the concept of a “**Vertical Urban Hub**”, where mobility, commerce, work, and living are integrated into one connected building.
- The design uses a flexible commercial layout that can adapt to different businesses and future needs. EV-ready parking, green balconies, shaded façades, and a landscaped roof introduce sustainable features into a compact urban structure.

1. TECHNOLOGIES TO BE USED :

- Autodesk Revit – Architectural, structural, floor plans, elevations, sections, and 3D building modelling.
- Autodesk 3D Visualisation/Rendering – Present the proposed building through realistic 3D views.
- Dynamo for Revit – Optional visual programming for automation, repetitive modelling, schedules, and design optimisation.
- AutoCAD – Optional preparation and reference of 2D drawings and site information.
- Excel – Area statements, room schedules, parking calculations, and quantity documentation.

2. METHODOLOGY AND PROCESS FOR IMPLEMENTATION:

Problem Analysis → Site & Requirement Study → Concept Development → Site Planning → Space Zoning → Structural Grid → Revit Modelling → Sustainability and climate Integration → 3D Visualization → Documentation → Final Presentation

1. POTENTIAL CHALLENGES AND RISKS:

- Limited site area may create difficulties in accommodating parking, circulation, and open spaces. Conflicts may occur between commercial and residential circulation.
- High-rise construction requires careful structural and service coordination.
- Basement parking and EV charging require proper fire and electrical planning.
- Maintaining natural lighting and ventilation in a dense urban building can be challenging.

2. STRATEGIES FOR OVERCOMING THESE CHALLENGES:

- Use efficient space planning and a compact structural grid to maximize usable area.
- Provide separate entrances and circulation cores for commercial and residential users..
- Allocate dedicated areas for EV charging, electrical systems, fire safety, and building services.
- Use balconies, shaded windows, courtyards/voids, and façade fins to improve daylight and ventilation.
- Introduce variations in balconies, materials, screens, and greenery to create a dynamic façade.

1. POTENTIAL IMPACT ON THE TARGET AUDIENCE :

- Residents benefit from comfortable housing, secure access, natural light, ventilation, and convenient access to nearby commercial facilities. Businesses gain strategically located commercial spaces with high visibility and accessibility.
- Visitors benefit from organized circulation, parking, and pedestrian-friendly spaces. The integrated development also creates a more connected and active urban environment by combining **living, working, shopping, and mobility** within one building.

2. BENEFITS OF THE SOLUTION :

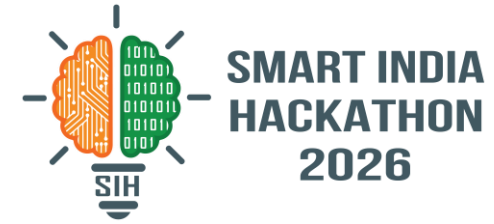
Economic Benefits:

Maximizes the value of limited urban land.
Generates income through commercial and residential spaces.
Supports local businesses and employment.

Social Benefits:

Creates a convenient live-work environment.
Provides safe and organized spaces for residents and visitors.
Encourages interaction through shared community areas.

RESEARCH AND REFERENCES



- **Mixed-Use Development & Sustainability — Nagpur, India**
[Research paper – Sustainability of Mixed-Use Neighbourhoods in Nagpur](#)
- **Vertical Mixed-Use Development**
[Research – Real Option and Vertical Mixed-Use Development](#)
- **Mixed-Use Development & Urban Sustainability**
[Research – Policy Coherence in Mixed-Use Development](#)
- **Mixed-Use Urban Form & Environmental Performance**
[Research – Mixed-Use and Open Urban Neighbourhoods](#)
- **National Building Code of India 2016**
[BIS – National Building Code of India 2016](#)
- **Autodesk Revit / BIM**
Autodesk also describes Revit workflows with Dynamo and sustainability analysis tools.
[Autodesk Revit for Architecture](#)